

Discussion 1: Introduction to Special Relativity – Solutions

Names:

Section:

Part 1: Galilean Transformations

Before moving on to the curious properties of relativity, we will first review how to describe physics in two different coordinate systems with relative velocity $v \ll c$. This goes by the fancy name of “Galilean transformations,” but these are just the ordinary Newtonian kinematics you are already familiar with from PHYS 211.

The setup is as follows: imagine you are standing next to a railroad track watching a train go by. Your reference frame will be called S , and in S , the train has velocity v in the $+x$ direction. The reference frame of a person on the train will be called S' . In all of the following, you are *strongly encouraged to draw pictures where appropriate!* Nothing in this part should require more than 2-3 lines of algebra. These are not trick questions of any kind, they are meant to get you thinking carefully about “simple” concepts like length and time in a precise way.

- 1.1. After a time Δt has elapsed, how far has the train moved in S ?

The train moves a distance $v\Delta t$.

- 1.2. Suppose a person on the train (let’s call them Alice) walks from the back of the train to the front, a distance $\Delta x'$ in S' , taking a time Δt (as measured on your watch) to do so. How far has Alice traveled in your frame during the time Δt , in terms of $\Delta x'$ and Δt ? Let’s call this distance Δx .

Alice moves $\Delta x'$ and the train she’s on moves $v\Delta t$, so $\Delta x = \Delta x' + v\Delta t$.

- 1.3. In Newtonian mechanics, we simply *assume* that the same amount of time elapses in both S and S' , $\Delta t' = \Delta t$. Using this fact, show that your result for problem 2 is equivalent to *Galilean transformations*

$$\Delta t = \Delta t', \tag{1}$$

$$\Delta x = \Delta x' + v\Delta t' \tag{2}$$

which allows you to express the coordinates $(\Delta t, \Delta x)$ of S solely in terms of the coordinates $(\Delta t', \Delta x')$ of S' .

Using the result from 1.2, but substituting $\Delta t'$ for Δt , gives Eq. (2).

- 1.4. Solve for $\Delta x'$ and $\Delta t'$ in terms of Δx and Δt to derive the *inverse transformations*, which lets you express quantities in S' in terms of those in S .

$\Delta x' = \Delta x - v\Delta t$, $\Delta t' = \Delta t$. This makes sense because in the frame of the train, the ground is moving backwards, with velocity $-v$.

- 1.5. The Δ 's are sometimes annoying, and we can get rid of them by defining the origins of the respective coordinate systems. Suppose we choose to define the origin of S' ($x' = y' = z' = 0$) to be the back of the train, the origin of S ($x = y = z = 0$) to be your location on the track, and $t = t' = 0$ to be the time at which the back of the train passes you. First, describe in a few sentences how you and Alice would perform this synchronization of your coordinate systems. You may assume you have access to any tools you need: watches, rulers, paintball guns, laser pointers, etc. Then, letting the total length of the train be L , use the Galilean transformations to determine x , the location in S of Alice once they have walked to the front of the train, in terms of v , t , and L . Your answer should have no Δ 's.

To synchronize the distances, Alice could place a ruler with the zero mark at the back of the train, and you could place a ruler with zero mark at your location. Alice attaches a paintbrush to the back of the train: the first drop of paint which appears on your ruler will coincide with both the origin of S and the origin of S' . To synchronize the times, suppose Alice's paintbrush has a piece of metal attached. At the moment that the paintbrush starts painting your ruler, the metal will close an electrical circuit that starts a stopwatch on both your platform and the train. In the limit where the length of your two circuits becomes infinitely small, your stopwatch and Alice's will start simultaneously. With everything synchronized, we have $\Delta x' = L$ and $\Delta t = t$, as well as $\Delta x = x$, so $x = vt + L$.

- 1.6. Assuming Alice walks with constant speed, what is their speed v_A in frame S' in terms of L and t ? Let their speed as seen in reference frame S be v_S . Use the Galilean transformations to derive the *velocity addition rule*,

$$v_S = v_A + v. \quad (3)$$

In S' , Alice's speed is simply $v_A = L/t$. In S , the observed speed is $v_S = x/t$. Dividing both sides of the solution to problem 1.5 by t gives $v_S = v + v_A$.

- 1.7. The velocity addition rule is actually much more general, and can be derived by taking the Δ 's in the Galilean transformations to be infinitesimal. Defining $v_A \equiv \Delta x' / \Delta t'$, show that you obtain the same velocity addition rule (3).

Going back to the Galilean transformations and dividing Eq. (2) by $\Delta t'$, we have $\Delta x / \Delta t' = \Delta x' / \Delta t' + v$. But since $\Delta t = \Delta t'$, the left-hand side is $\Delta x / \Delta t \equiv v_S$, so $v_S = v_A + v$.

- 1.8. The reason that we focus on systems moving at constant velocity (called *inertial frames*) is that Newton's Laws imply the *principle of Galilean relativity*: the Newtonian laws of physics are the same in any two inertial frames. Using the fact that acceleration is the time derivative of velocity, $a = dv/dt$, show that the velocity addition rule implies that any acceleration measured in S is the same as that measured in S' . (Hint: take a time derivative of both sides of the velocity addition rule.) By Newton's second law, $F = ma$, this means that forces measured in both frames are guaranteed to be the same.

Taking time derivatives gives $dv_S/dt = dv_A/dt + dv/dt$. For an inertial system, $dv/dt = 0$ since v is constant. dv_S/dt is simply a_S , which is the acceleration in S , and (using $dt = dt'$) dv_A/dt is $a_{S'}$. Thus $a_S = a_{S'}$.

Part 2: Constant speed of light and loss of simultaneity

Essentially all of the counterintuitive properties of relativity arise from the postulate that *the speed of light in vacuum is the same in all inertial reference frames*, $c = 2.99792458 \times 10^8$ m/s.¹ Here you will investigate the first such weirdness, the loss of simultaneity; we will meet the other strange relativity effects next week. We will consider the same setup as in Part 1, with S and S' being the ground and train frames, respectively. Let L be the length of the train as measured in S (we will see in the following weeks the importance of this somewhat pedantic distinction). *Draw pictures!*

Note: this part of the discussion is essentially a rephrasing of Section 1.3.1 in Morin

- 2.1. At $t = t' = 0$, Alice (standing at position $x' = 0$) shines a laser pointer toward the front of the train. Argue that in S' , the position of the light pulse emitted from the laser pointer is given by $x' = ct'$.

Since speed in S' is $\Delta x' / \Delta t'$ (see 1.7 above), and light is assumed to have speed c in S' , we have $x' = ct'$.

- 2.2. Assuming the constancy of the speed of light, we *must* have that in S , the position of the light pulse is given by $x = ct$. Argue that it is *impossible* to have $t = t'$ when $v \neq 0$. If the speed of light is truly constant in any inertial frame, then the Galilean assumption $\Delta t = \Delta t'$ cannot be correct!

By the same logic as above, the speed of light being c in S means we must have $x/t = c$, or $x = ct$. If we were to have $t = t'$, this would mean $ct = ct'$, implying $x = x'$, which is only possible if the train is not moving ($v = 0$)!

- 2.3. We can make this *loss of simultaneity* even sharper by showing that events which are simultaneous in S' (i.e. happen at the same time t') are not simultaneous in S . Specifically, suppose that at $t = t' = 0$, Alice stands exactly at the middle of the train and shines two laser pointers F and B towards the front and the back of the train, respectively. Clearly, in S' the light pulses will hit the ends of the train at the same time: in other words, these events are *simultaneous*. Use the fact that both the back and front of the train are moving with speed v in S to show that, as measured in S , the times t_F and t_B at which the light pulses hit the front and back of the train are

$$t_F = \frac{L}{2(c-v)}, \quad t_B = \frac{L}{2(c+v)}. \quad (4)$$

These times are not equal! The same events, which are simultaneous in S' , are *not simultaneous* in S . So the question “do the light pulses hit the front and back of the train at the same time?” does not have an unambiguous answer – it depends on your reference frame!

After a time t (measured in S), the back of the train has moved a distance vt (from 1.1), while the light has traveled a distance $\Delta x = ct$. If the light (which starts at position $L/2$ in S , in the middle of the train)² is traveling in the same direction as the train, then after a

¹Note that’s an “=” not “ $\approx$ ”: the speed of light in vacuum is used to *define* the meter, so that relation is *exact*!

²While we’re questioning all of our assumptions, you might wonder: just because Alice is in the middle of the train in S' , how do we know she is *also* in the middle of the train in S ? It turns out that this is required by the assumption of *linearity* of coordinate transformations which preserve the laws of physics in inertial frames, which we will get to in week 3.

time t , the light beam is at $L/2 + ct$ while the front of the train is at $L + vt$ (all distances are measured in S). The time when these distances coincide defines t_F , so

$$L/2 + ct_F = L + vt_F \implies t_F = \frac{L}{2(c - v)}.$$

Similarly, when the light is traveling opposite the direction of the train, the light beam is at $L/2 - ct$ while the back of the train is at vt , so

$$L/2 - ct_B = vt_B \implies t_B = \frac{L}{2(c + v)}.$$

Part 3: Relativistic velocity addition

Among the many things which must be modified as a result of the postulate of the constancy of the speed of light is the Galilean velocity addition rule you derived in Part 1. For now, we will just state the result, which you will derive later this semester. If an object A moves with velocity v_A in S' , and S' is moving with velocity v with respect to S , then the velocity v_S of A in S is given by the *relativistic velocity addition rule*

$$v_S = \frac{v_A + v}{1 + v_A v / c^2}. \quad (5)$$

- 3.1. If $v_A \ll c$ (for example, a person walking as in Part 1, or a ball thrown out the train window), show that the relativistic velocity addition rule reduces to $v_S \approx v_A + v$, which is the Galilean result. So Galilean transformations are fine as long as all speeds are small compared to c .

When $v_A \ll c$, we can approximate the denominator as $1 + v_A v / c^2 \approx 1$. Keeping only the numerator gives $v_S = v_A + v$, the Galilean velocity addition rule.

- 3.2. If $v_A = c$ (for example, a laser pointer as in Part 2), show that $v_S = v_A$. The relativistic velocity addition rule is consistent with the constancy of the speed of light! In particular, the speed of light in S is *not* $v + c$; light does not get “carried along with the train” at all.

Plugging in $v_A = c$ gives

$$v_S = \frac{c + v}{1 + cv/c^2} = \frac{c + v}{1 + v/c} = \frac{c(c + v)}{c + v} = c.$$

- 3.3. If $v = 0.5c$, how large does v_A have to be for v_S to be smaller than its Galilean value by 1%? How long (as measured in S) would it take an object moving at this speed v_A to get from Earth to the Moon, a distance of 3.84×10^5 km? This should give you some sense for just how *fast* things must be moving to see any deviation from the Galilean transformation rules!

The Galilean value is $v_S = v_A + v = v_A + 0.5c$. So we want to solve

$$\frac{v_A + 0.5c}{1 + 0.5v_A/c} = 0.99(v_A + 0.5c)$$

$$\frac{1}{1 + 0.5v_A/c} = 0.99$$

$$0.01 = 0.495v_A/c$$

$$v_A = 0.02c \approx 6 \times 10^3 \text{ km/s}$$

At this speed, the Earth-Moon trip would take about a minute!